Abhishek Deelip Yawalkar

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SUMMARY

AI Software Engineer with hands-on experience in Generative AI evaluation at Outlier and building end-to-end AI applications. Skilled in RAG, agentic workflows, semantic search, and knowledge graphs, with projects in document intelligence and supplier risk analysis. Recognized for high-quality engineering and awarded 2nd place at CS Demo Day for a mobile disease-detection application.

TECHNICAL SKILLS

AI & MLOps: LangGraph, LangChain, RAG Pipelines, Knowledge Graphs, GraphRAG, Ontology Modeling, Vector Search (pgvector), Agentic Workflows, Prompt Engineering
Programming Languages: Python, TypeScript, JavaScript, Java, SQL
Backend & APIs: FastAPI, Node.js, Flask, RESTful APIs
Frontend: React.js, React Native, Tailwind CSS, Zustand, AngularJS, HTML5/CSS3
Databases & Data Processing: PostgreSQL, MongoDB, MySQL, SQLAlchemy, Neo4j, Pandas, RapidFuzz
Cloud, DevOps & Tools: Docker, AWS, GitHub Actions, CI/CD, Nginx, Git, GitHub, GitLab
Testing: PyTest, Jest, Unit & Integration Testing

WORK EXPERIENCE

AI Software Engineer (Independent Contractor, Project-Based) July 2024 - June 2026
Outlier - Canada, Quality Ratings - 5/5

- Boosted AI model accuracy from 73% to 97% by developing coding prompts and evaluating generated code across Python, Java, TypeScript, and SQL against internal quality standards.
- Contributed to a reported 20% reduction in AI-generated code defects, as documented in internal evaluation reports, by conducting security audits and peer reviews.
- Strengthened AI training data quality by refining peer-submitted code for correctness, efficiency, and maintainability to meet internal evaluation standards.

PROJECTS

DocuMind: PDF Intelligence System August 2026 - October 2026

- Built an agentic document processing system using a 5-node LangGraph workflow and self-hosted local LLMs (Ollama) to automate PDF summarization, entity extraction, and GDPR/PII compliance scanning without external API costs.
- Implemented RAG-based question answering with PostgreSQL and pgvector, enabling semantic retrieval and natural-language queries across multi-page documents.
- Developed and containerized a full-stack application using FastAPI, React, and TypeScript, with GitHub Actions CI/CD for automated testing and Docker build validation.

Supplier Risk Graph July 2026 - August 2026

- Identified potential supplier risk exposure across 500+ corporate records by building an ontology-driven Neo4j knowledge graph that resolved entities and traced ownership relationships up to 4 hops.
- Achieved 85% retrieval accuracy across 20 benchmark tests by integrating semantic search with Cypher-based graph traversal in a hybrid GraphRAG pipeline.
- Enabled analysis of 20+ supplier risk scenarios through an interactive FastAPI and Streamlit application that visualizes ownership paths and explains graph-based risk connections.

Tomato Crop Disease Identification January 2024 - April 2024
CS Demo Day - 2nd Place Winner

- Achieved 94% accuracy in detecting crop leaf diseases by developing and training a CNN deep learning model on localized computer vision datasets.
- Led a 4-engineer team to design and deploy a scalable, high-availability backend microservice architecture on Google Cloud Platform (GCP).

- Built cross-platform iOS/Android interfaces in React Native, optimizing mobile asset delivery and image payload processing for instant on-device classification.

VOLUNTEER EXPERIENCE

IT Assistant February 2024 - June 2024
Windsor Food Bank, Canada

- Automated inventory tracking and volunteer scheduling systems using Python scripts and Google Sheets, eliminating manual data entry errors and cutting administrative overhead.
- Engineered a web-based donation portal with custom HTML/CSS integration, centralizing donor intake and streamlining data collection workflows.

EDUCATION

Master of Applied Computing January 2023 - April 2024
University of Windsor, Canada

- Key coursework: Advanced Software Engineering Topics, Advanced Database Topics, Advanced Systems Programming, Internet Application or Distributed Systems, Network and Data Security

Master of Science in Information Technology (Cloud Computing) May 2019 - June 2021
Mulund College of Commerce, India

- Key coursework: Microservices Architecture, Cloud Application Development, Data Science

Bachelor of Science in Computer Science April 2016 - April 2019
Mulund College of Commerce, India

- Key coursework: Core Java, Advanced Java, Data Structures and Algorithms, Software Engineering, Software Testing, Database Management System, Web Development

CERTIFICATES & TECHNICAL TRAINING

- Forage - Advanced Software Engineering Job Simulation, Walmart Global Tech
- AI Fluency: Framework & Foundations, Anthropic
- Getting Started with Generative AI, IBM SkillsBuild
- Software Design: From Requirements to Release, LinkedIn Learning